

MULTIMODAL AI FRAMEWORK FOR EARLY PARKINSON'S DISEASE DETECTION

RP 25-26J-109

Project Proposal Report

Atigalage Y. V

B.Sc. (Hons) Degree in Information Technology, Specialized in Data
Science

Department of Computer Science and Software Engineering
Sri Lanka Institute of Information Technology
Sri Lanka

August 2025

SPEECH & VOCAL BIOMARKER ANALYSIS FOR EARLY DETECTION OF PARKINSON'S DISEASE

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DECLARATION

I declare that this is my own work, and this proposal does not incorporate without acknowledgement any material previously submitted for a degree or diploma in any other university or Institute of higher learning and to the best of our knowledge and belief it does not contain any material previously published or written by another person except where the acknowledgement is made in the text.

Name	Student ID	Signature
Atigalage Y.V	IT22079336	

The above candidates are carrying out research for the undergraduate Dissertation under my supervision.

Signature of the supervisor:



Date: 2025-08-31

ABSTRACT

Parkinson's Disease (PD) is the fastest-growing neurodegenerative disorder worldwide, with early signs often emerging as subtle vocal changes such as increased jitter, shimmer, monotone speech, and irregular pauses. Current diagnostic approaches largely rely on clinical evaluations that detect PD only after considerable neurological damage, delaying timely intervention.

This research focuses on developing a non-invasive, AI-based method to identify early-stage PD through speech and vocal biomarker analysis. Using publicly available datasets such as the Parkinson's Voice Initiative, voice recordings will be preprocessed to remove background noise and extract key acoustic features like Mel-frequency cepstral coefficients (MFCCs), pitch variation, jitter, and shimmer as well as linguistic features including pause frequency and verbal fluency. Feature extraction will be performed using signal processing and Natural Language Processing (NLP) techniques.

Several machine learning models including Support Vector Machines (SVM), Long Short-Term Memory (LSTM) networks, and ensemble classifiers will be trained and evaluated to classify speech patterns affected by PD. To improve clinical interpretability, Explainable AI (XAI) methods such as SHapley Additive exPlanations (SHAP) and attention mechanisms will be integrated to highlight influential features in model predictions.

The expected outcome is an accurate AI model deployed via a user-friendly mobile application for real-time, scalable screening. This tool will also contribute to a broader multimodal diagnostic framework that integrates wearable sensor data, drawing analysis, and digital cognitive tests. Ultimately, this work aims to improve early PD detection, enabling earlier interventions and better patient outcomes.

Keywords: Parkinson's Disease, Speech Analysis, Machine Learning, Explainable AI, Early Detection, Natural Language Processing

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LIST OF ABBREVIATIONS

Abbreviation	Description
PD	Parkinson's Disease
AI	Artificial Intelligence
ML	Machine Learning
DL	Deep Learning
NLP	Natural Language Processing
XAI	Explainable AI
SHAP	SHapley Additive exPlanations
LSTM	Long Short Term Memory
CNN	Convolutional Neural Network
MFCC	Mel-Frequency Cepstral Coefficients
SVM	Support Vector Machine

1.INTRODUCTION

Parkinson's Disease (PD) represents a significant neurodegenerative disorder characterized by progressive deterioration of motor functions and speech capabilities. Traditional diagnostic approaches, which predominantly rely on clinical observation of motor symptoms, frequently fail to identify the condition during its early stages when therapeutic interventions could be most beneficial. This diagnostic limitation stems from the fact that visible motor manifestations typically emerge only after substantial neural degeneration has occurred. Recent studies, however, have demonstrated that subtle alterations in speech patterns, including variations in vocal stability, articulation, and prosody, may serve as potential prodromal indicators, appearing several years before the onset of characteristic motor symptoms.

The present research investigates an innovative computational methodology for early PD detection through systematic analysis of vocal characteristics. Voice recordings from established clinical datasets undergo comprehensive signal processing to quantify acoustic parameters, including jitter, shimmer, and spectral features. Concurrently, nlp techniques are employed to evaluate linguistic patterns, including speech fluency, pause dynamics, and verbal expression. These multidimensional vocal biomarkers are subsequently analyzed using machine learning algorithms specifically optimized for binary classification of neurological health status. The methodology incorporates rigorous validation protocols, utilizing both cross-validation techniques and independent test datasets to ensure robust performance evaluation across diverse demographic groups.

A distinctive aspect of this investigation involves the integration of explainable AI techniques to enhance clinical interpretability. By implementing feature importance analysis and decision visualization methods, the classification outputs maintain transparency, thereby facilitating potential adoption within medical diagnostic workflows. The proposed system architecture is designed with scalability in mind, potentially enabling deployment as a mobile health application for widespread screening applications, particularly in resource-constrained healthcare environments.

Special consideration is given to optimizing computational efficiency to ensure practical implementation on consumer-grade mobile devices without compromising diagnostic accuracy.

This research constitutes one component of a broader multimodal diagnostic framework under development. When combined with complementary digital biomarkers derived from wearable motion sensors and computerized drawing analysis, the vocal analysis component contributes to a more comprehensive neurological assessment paradigm. Emphasis is placed on computational efficiency and data privacy considerations throughout the system design process, ensuring both technical robustness and ethical compliance. The framework incorporates stringent data anonymization protocols and secure transmission mechanisms to protect sensitive health information while maintaining diagnostic efficacy.

The potential clinical implications of this work are considerable. By establishing vocal characteristics as reliable digital biomarkers for PD, this research may contribute to earlier diagnostic capabilities, more timely therapeutic interventions, and ultimately improved patient outcomes. The integration of advanced computational techniques with clinical neuroscience represents a promising avenue for transforming traditional diagnostic paradigms into neurodegenerative disease management. Future research directions include longitudinal studies to validate the predictive value of vocal biomarkers and investigations into their potential utility for monitoring disease progression and treatment response.

1.1 Background literature and survey

Parkinson's Disease (PD) is a progressive neurodegenerative disorder and the second most common after Alzheimer's disease, affecting nearly 1% of individuals over the age of 60 worldwide [1]. It results from the gradual degeneration of dopaminergic neurons in the substantia nigra, a midbrain region essential for motor control. This loss leads to motor symptoms such as tremors, rigidity, and bradykinesia, as well as non-motor symptoms including sleep disturbances, depression, and changes in speech. Traditionally, diagnosis relied on the clinical observation of motor symptoms. However, evidence increasingly indicates that subtle non-motor signs, particularly speech and voice abnormalities, may appear several years before motor symptoms become clinically apparent [2]. Identifying these early markers offers an opportunity for timely diagnosis and intervention, which can improve quality of life and enable earlier treatment planning. Detecting PD at this pre-motor stage also opens possibilities for preventative strategies and personalized care pathways before irreversible neural damage occurs.

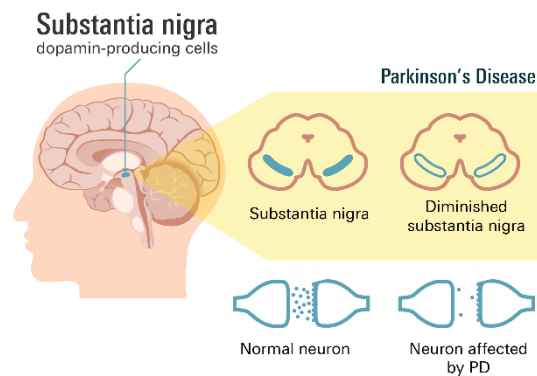


Figure 1: Degeneration of the substantia nigra

Speech analysis has gained attention as a promising, non-invasive, and cost-effective tool for early PD detection. Nearly 90% of individuals with PD develop speech impairments collectively known as hypokinetic dysarthria [3]. This condition is characterized by reduced vocal loudness, monotonic pitch, imprecise articulation, and irregular rhythm. These impairments arise from PD's disruption of basal ganglia circuits, which play a critical role in motor planning and coordination, including fine control of vocal tract muscles [4]. Early studies systematically described these deficits,

providing a foundation for quantitative assessment approaches [5]. Their work inspired decades of research into developing objective, repeatable measures of speech deterioration in neurological disorders.

Advances in digital signal processing have made it possible to capture minute variations in speech that are imperceptible to the human ear. Tsanas et al., for instance, achieved high classification accuracy by extracting non-linear features such as jitter (pitch variation), shimmer (amplitude variation), and harmonic-to-noise ratio from sustained vowel sounds [6]. These parameters reflect vocal fold vibration stability, which is often disrupted in PD. Little et al. expanded this approach by analyzing connected speech, including reading and spontaneous conversation, showing that natural speech offers richer diagnostic cues than isolated vowel phonation [7]. Large datasets like the Parkinson's Voice Initiative, containing thousands of telephone-collected voice samples, have accelerated research by providing standardized benchmarks for evaluation [6], [7]. Such large-scale resources are critical for validating algorithms across diverse populations and ensuring their clinical applicability.

Researchers have also recognized the value of multidimensional speech analysis. Rusz et al. proposed a framework incorporating phonation, articulation, prosody, and fluency to detect early PD and track progression [8]. More recently, multimodal approaches have integrated acoustic features with articulatory movement data and cognitive-linguistic markers. Vasquez-Correa et al., for example, combined acoustic, articulatory, and linguistic features to improve classification accuracy in distinguishing PD from healthy controls [9]. These integrated models provide a more holistic view of how PD impacts both the motor and cognitive-linguistic aspects of speech, offering richer diagnostic insights.

In addition, NLP has emerged as a valuable approach for examining higher-level linguistic patterns. PD can subtly affect word choice, sentence structure, and information density in speech, often reflecting cognitive-linguistic decline. NLP techniques can quantify these patterns, identifying reduced syntactic complexity, limited vocabulary diversity, and semantic drift. Studies show that narrative speech

tasks, such as describing a picture or telling a story, can achieve high classification accuracy using linguistic features alone [8]. Transformer-based models like BERT have further improved detection by capturing subtle semantic and contextual changes missed by traditional NLP methods. These models allow for a deeper understanding of how PD affects natural communication beyond purely acoustic changes.

Machine learning is central to converting extracted features into reliable predictions. Early research often used traditional classifiers such as Support Vector Machines (SVMs) and Random Forests trained on handcrafted features [6], [9]. With the rise of deep learning, CNNs have been used to process raw audio directly, automatically learning feature representations. Hybrid models that combine deep learning feature extraction with classical machine learning classifiers have shown potential for improving both performance and interpretability.

As model complexity increases, explainability becomes crucial for clinical adoption. Clinicians are more likely to trust AI systems if they can understand the reasoning behind predictions. Explainable AI (XAI) techniques, such as SHAP (Shapley Additive exPlanations), highlight key features such as increased jitter or reduced pitch variation that drive predictions, supporting transparency [10]. Interpretable models like decision trees, when based on physiologically relevant features, remain valuable where simplicity is a priority. Importantly, these interpretable approaches help bridge the gap between advanced computational methods and clinical decision-making.

Technological advances are also making PD speech analysis more practical. Mobile devices and wearables can now run PD detection algorithms, enabling remote monitoring. Smartphone applications can record and analyze short speech samples, while smartwatches equipped with microphones can passively track vocal changes. TinyML, which enables efficient machine learning on embedded devices, allows lightweight CNN models to run entirely on-device, enabling real-time processing and preserving privacy [9]. This portability and efficiency make such tools ideal for deployment in low-resource healthcare settings or for continuous at-home monitoring.

Nonetheless, several challenges remain. Many studies are based on small or demographically narrow datasets, limiting generalizability. Variations in recording

conditions, such as microphone quality, background noise, and speaking style, can affect performance. Moreover, some speech patterns associated with PD are not unique to the disease, overlapping with other movements or neurological disorders. Initiatives like the Parkinson's Progression Markers Initiative (PPMI) are addressing these limitations by collecting standardized, longitudinal speech and clinical data from a diverse, global cohort [8].

Future research may focus on integrating speech biomarkers with other digital health measures, such as gait, handwriting, and eye-tracking, to improve diagnostic specificity. Multimodal AI systems could provide clinicians with richer, cross-validated evidence for diagnosis and monitoring, potentially enabling earlier intervention and personalized treatment.

In conclusion, current research trends involve combining acoustic and linguistic features with advanced machine learning and NLP, while prioritizing interpretability and computational efficiency. The use of XAI ensures clinically meaningful explanations alongside accurate predictions. Models optimized for mobile deployment could facilitate large-scale screening, especially in areas with limited neurological expertise. With robust validation across diverse datasets, these systems hold promises for becoming accessible, scalable tools for early PD detection, advancing the role of AI-assisted healthcare in clinical settings.

1.2 Research Gap

Recent advancements in speech-based Parkinson's Disease (PD) detection have demonstrated the potential of machine learning (ML) and deep learning (DL) for identifying early disease markers. However, analysis of key studies reveals several limitations that hinder their translation into practical, real-world screening tools.

Tsanas et al. (2010) [6] applied ML techniques to extract and analyze acoustic features such as MFCC, jitter, and shimmer from sustained phonation recordings. While their approach achieved high classification accuracy, it focused solely on acoustic parameters and lacked integration of linguistic or cognitive-linguistic markers. Furthermore, the study did not incorporate deep learning, explainable AI, or real-time deployment, limiting its scalability and clinical interpretability.

Vasquez-Correa et al. (2019) [8] introduced a multimodal approach that combined acoustic, articulatory, and linguistic features using deep learning models. This broadened the diagnostic scope and improved classification performance. However, the system relied on computationally intensive models unsuitable for resource-constrained environments. Additionally, it lacked explainable AI components, making it difficult for clinicians to interpret predictions and trust the system's decision-making process.

Rusz et al. (2021) [2] focused on identifying prodromal PD markers by examining acoustic and linguistic patterns before the onset of motor symptoms. Their findings highlighted valuable early indicators of PD, yet the study did not leverage deep learning or explainable AI and was not designed for real-time or mobile deployment.

From these analyses, three critical gaps emerge:

1. **Absence of Explainable AI in Speech-Based PD Detection**

None of the reviewed studies incorporated SHAP, attention mechanisms, or similar interpretability tools. Without explainability, clinical adoption is hindered, as healthcare professionals require transparent reasoning behind automated predictions.

2. Lack of Real-Time, On-Device Processing

Existing Research Often relies on cloud-based or high-performance computing resources, limiting deployment in low-resource or remote healthcare settings. Real-time, on-device analysis using frameworks like TinyML remains largely unexplored in PD speech analysis.

3. Limited Mobile Accessibility for Widespread Screening

No reviewed study offered a fully deployable mobile solution capable of recording, processing, and interpreting speech locally. Mobile-enabled screening tools could significantly expand access to early diagnosis, especially in underserved regions.

The proposed system directly addresses these gaps by integrating acoustic and linguistic feature analysis, combining ML and DL models, embedding explainable AI, and deploying the framework on mobile devices with real-time on-device processing. This approach aims to deliver a scalable, interpretable, and privacy-preserving solution for early PD detection. The following table shows the summary of the above gap analysis for further understanding.

Features	Tsanas et al. (2010) [6]	Vasquez-Correa et al. (2019) [8]	Rusz et al. (2021) [2]	Proposed System
Use of Machine Learning	✓	✓	✓	✓
Use of Deep Learning	✗	✓	✗	✓
Acoustic Feature Analysis (e.g., MFCC, jitter, shimmer)	✓	✓	✓	✓
Explainable AI (SHAP, Attention)	✗	✗	✗	✓
Real-Time / On-Device Processing	✗	✗	✗	✓
Mobile Deployment	✗	✗	✗	✓

Table 1: Research Gap

1.3 Research Problem

Parkinson’s Disease (PD) remains challenging to detect in its early stages, as current diagnosis typically depends on visible motor symptoms that only emerge after considerable neurodegeneration. Existing clinical tools, including neurological examinations and DaTscan imaging, are either subjective, costly, or inaccessible for routine screening. As a result, diagnosis is often delayed by 5–10 years from disease onset, reducing the window for effective neuroprotective interventions and increasing long-term healthcare costs.

Emerging evidence shows that PD often causes subtle but distinct vocal changes such as reduced loudness, monotonic speech, and articulatory imprecision, many years before motor symptoms appear. Voice analysis, therefore, presents a promising, non-invasive method for early screening. However, several barriers still prevent its clinical adoption. First, PD influences both the physical production of speech (acoustic features) and the way language is used (linguistic patterns), yet most current methods examine only one dimension. Second, the lack of standardized protocols across languages and dialects limits the generalizability of findings. Third, many systems are not designed for interpretability or integration into everyday healthcare workflows, especially in resource-limited settings.

Complicating matters further, PD presents differently from one person to another. Vocal symptoms vary in both type and severity, while natural age-related changes in voice can obscure early signs of the disease. Advanced analytical techniques are needed to distinguish pathological alterations from normal variations. Despite mounting evidence of their value, voice-based biomarkers have not yet been incorporated into official diagnostic criteria.

A practical solution would involve a vocal analysis system that:

1. Extracts comprehensive acoustic and linguistic features.
2. Uses explainable AI methods, such as SHAP values and attention mechanisms, for clinically interpretable outputs.
3. Supports mobile-optimized deployment for accessible screening in both clinical and community settings.

Such a system could enable accessible early detection through routine voice recordings, support continuous disease monitoring, and improve treatment evaluation, transforming PD care across diverse healthcare settings.

2. OBJECTIVES

2.1 Main Objectives

This research aims to develop an AI-powered speech analysis system for early detection of Parkinson's Disease (PD) by identifying subtle vocal biomarkers. The system will analyze both acoustic features (jitter, shimmer, pitch variability) and linguistic patterns (pause frequency, speech rhythm) using advanced signal processing and machine learning techniques. By creating an interpretable, mobile-compatible solution, this component will contribute to a comprehensive multimodal framework for scalable, accurate, and non-invasive PD detection.

2.2 Specific Objectives

The speech analysis component will achieve its main objective through these focused technical implementations:

- **To extract fine-grained vocal biomarkers from speech tasks**
Derive comprehensive acoustic and linguistic features from publicly available datasets (e.g., Parkinson's Voice Initiative, mPower) including jitter, shimmer, formant dynamics, pause patterns. These fine-grained vocal biomarkers detect subtle speech motor changes preceding visible Parkinson's symptoms, enabling sensitive early detection.
- **To develop predictive machine learning models for speech decline in PD**
Develop various machine learning classifiers including SVM, LSTM, and XGBoost tailored for early-stage Parkinson's detection. Train and fine-tune the models using extracted speech features. Employ cross-validation and hyperparameter tuning to maximize accuracy, targeting clinically relevant benchmark metrics by the sixth month of the project.
- **To integrate Explainable AI and validate clinically**
Integrate advanced explainability techniques like SHAP values and attention mechanisms to provide transparent model predictions. Correlate model outputs with recognized clinical speech evaluation tools such as the MDS-UPDRS.

Continuously validate and refine model reliability and clinical relevance throughout the project timeline to ensure practical applicability.

- **To validate and benchmark the speech analysis framework against established baselines**

Conduct rigorous validation using benchmark speech datasets and clinical speech evaluation scores (e.g., MDS-UPDRS). Compare model performance and interpretability with existing vocal biomarker detection frameworks to demonstrate improvements and added clinical value.

- **To integrate the speech module into the multi-modal mobile application**

Embed the vocal biomarker analysis framework into the group's multi-modal Parkinson's detection mobile app with real-time processing capabilities. This integration will enable the fusion of speech signals with other biomarkers (cognitive, motion, drawing), providing a comprehensive, scalable, and clinically relevant early detection system with user-friendly result visualization and feedback mechanisms.

3.METHODOLOGY

3.1 System Architecture Diagram

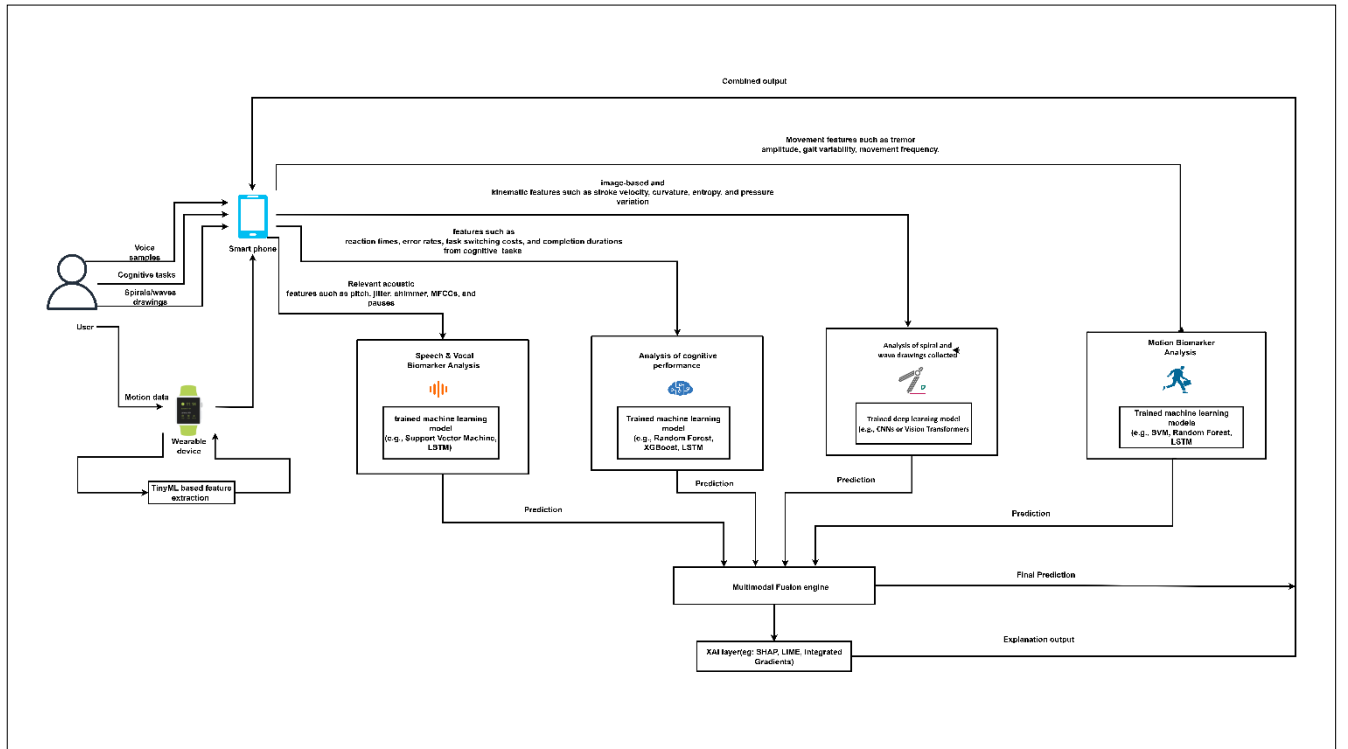


Figure 2 :System Architecture Diagram

This project presents a multimodal, AI-driven diagnostic system for the early identification of neurodegenerative disease Parkinson's. Four main data streams are integrated into the solution: wearable sensor data, spiral and wave drawing analysis, speech and vocal biomarkers, and digital cognitive testing using tasks related to language, memory, and executive function.

A dedicated module processes each input modality. To identify pauses, slurring, and monotony, speech signals are analyzed. Wearable sensors record tremors and abnormalities in movement. Image-based analysis is used to evaluate drawing tasks for motor abnormalities. Digital cognitive tests use task-based interaction data and NLP techniques to evaluate verbal fluency, recall, attention, and problem-solving.

To identify early cognitive decline, the extracted features are combined in a Multi-Modal Prediction Engine that employs ensemble models and deep learning.

TinyML-optimized lightweight models enable on-device deployment, guaranteeing real-time analysis while protecting user privacy. To improve trust and adoption, explainable AI (XAI) tools like attention maps and SHAP are integrated to make model decisions understandable to patients and clinics.

3.2 Component Architecture Diagram

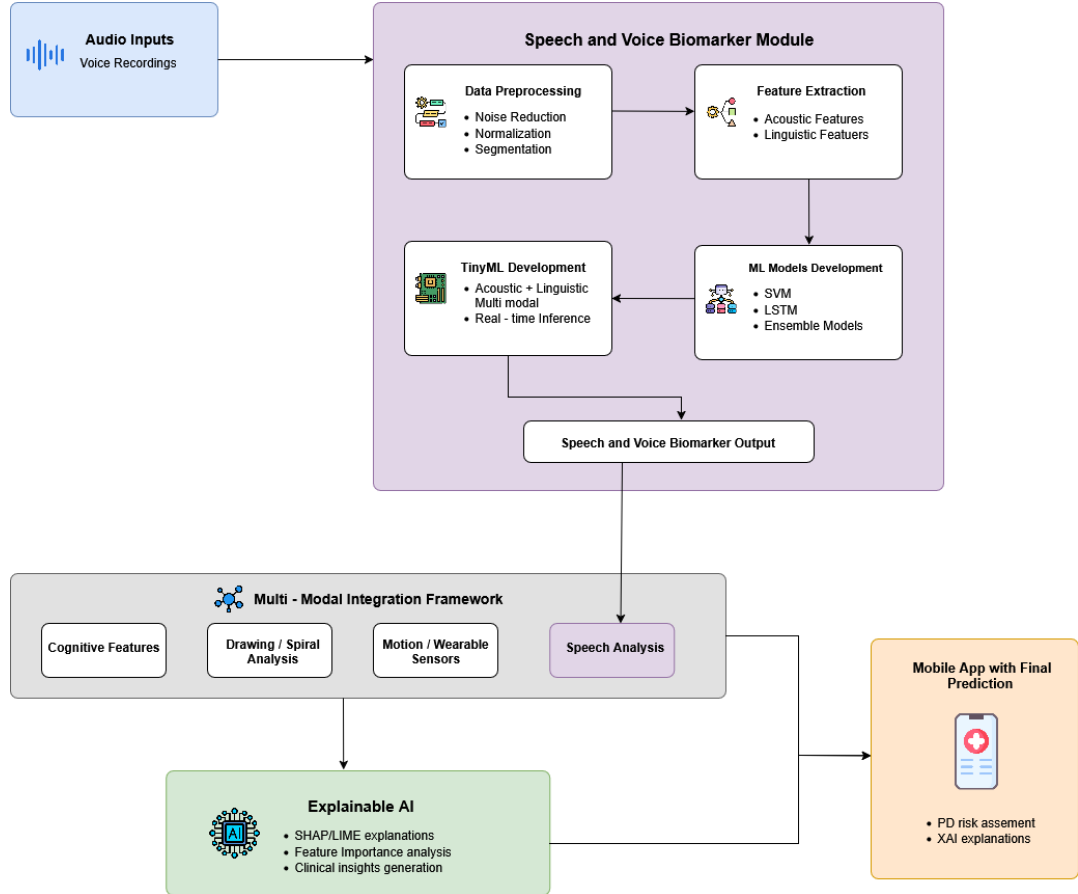


Figure 3 : Component Architecture Diagram

This research follows a structured five-phase approach to design a mobile-based AI system aimed at early detection of Parkinson’s Disease through speech analysis. The methodology combines techniques from signal processing, machine learning, and mobile app development to develop a reliable, accessible, and efficient diagnostic tool that operates directly on handheld devices.

1. Data Collection

Collect voice recordings from established, publicly available datasets such as the Parkinson's Voice Initiative, encompassing a variety of speech types including sustained vowels, scripted reading tasks, and natural spontaneous speech. This diversity ensures comprehensive capture of vocal characteristics relevant to Parkinson's detection.

2. Signal Preprocessing and Feature Extraction

Apply advanced signal processing techniques to clean raw audio data by reducing background noise and normalizing the volume levels. Segment the speech into analyzable units, then extract key acoustic features like jitter, shimmer, pitch variance, and MFCCs. Additionally, use natural language processing (NLP) to quantify linguistic features such as pause durations and verbal fluency. These modules will be optimized to run efficiently on TinyML platforms suitable for real-time deployment on mobile or wearable devices.

3. Model Development and Training

Develop robust machine learning models including Support Vector Machines (SVM), Long Short-Term Memory (LSTM) networks, and XGBoost ensemble classifiers to categorize speech samples as Parkinson's or healthy. Utilize cross-validation techniques on labeled datasets to ensure model generalizability. Tune hyperparameters systematically to maximize predictive accuracy. Evaluate model performance against clinical speech assessments such as the MDS-UPDRS speech sub-scores, using metrics including accuracy, precision, recall, F1-score, and ROC-AUC.

4. Explainability

Integrate interpretable AI techniques such as SHapley Additive exPlanations (SHAP) and attention mechanism visualizations to elucidate which acoustic and linguistic features most influence prediction outcomes. Develop user-friendly visualization dashboards that communicate these insights clearly, enabling clinicians and patients to understand the basis of each classification, thereby increasing trust and clinical acceptance of the AI system.

5. Integration and Deployment

Design and implement secure APIs to connect on-device TinyML feature extraction modules with a centralized cloud-based multimodal prediction engine, enabling seamless aggregation of data from speech, wearable sensors, and other modalities. Develop an intuitive mobile application interface that facilitates real-time patient interaction, remote monitoring, and feedback display, supporting continuous and accessible early Parkinson's screening.

3.3 Tools and Technologies

Core Implementation Language	 <i>Figure 4:Python</i>
Development Environment	 <i>Figure 6:Google Collab</i>  <i>Figure 5:VS Code</i>
Version Control	 <i>Figure 7:GitHub</i>
Design	 <i>Figure 8:Figma</i>
Mobile App Development	 <i>Figure 9:React Native</i>
Cloud / Backend Services	 <i>Figure 10 : Firebase</i>

Table 2:Tools and Technologies

4.PROJECT REQUIREMENTS AND DESIGN

4.1 Functional requirements

- Allow users to register, log in, and manage their profiles securely within the system.
- The system records various voice samples including sustained vowels, reading passages, and free speech to capture comprehensive vocal patterns.
- Perform efficient noise filtering and extract relevant speech features locally on the device.
- Classify Parkinson's disease probability using integrated multimodal biomarker data from various sensors.
- Provide clear and interpretable explanations for model predictions to support clinical decisions.
- Present analysis results through an intuitive and user-friendly interface on the mobile application.

4.2 Non-functional requirements

1. Accuracy & Reliability

The system needs to demonstrate strong accuracy in identifying Parkinson's cases when tested against well-established clinical datasets, ensuring doctors can trust it as a reliable diagnostic tool. It should work consistently well for people of all ages and from different linguistic backgrounds, making sure everyone receives fair access to quality healthcare screening regardless of their personal characteristics.

2. Speed & Efficiency

The system should analyze voice recordings rapidly enough to give healthcare providers immediate results that can inform their clinical decisions on the spot. It needs to run smoothly on everyday smartphones without requiring expensive, high-end devices, making it accessible to patients who use standard mobile technology.

3. Privacy & Security

The system must carefully protect patient privacy by following strict healthcare regulations like HIPAA and GDPR, ensuring that sensitive voice recordings never leave the patient's device unnecessarily. By processing everything locally on the

phone, it minimizes the risk of personal health information being intercepted or misused during transmission.

4. Usability & Accessibility

The system should be designed with elderly patients in mind, featuring easy-to-see buttons and straightforward instructions that accommodate people who may have difficulty with fine motor control or vision challenges. It needs to work in multiple languages so that patients from different cultural backgrounds can use it comfortably in their native tongue.

5. Scalability

The system should be robust enough to handle busy hospital environments where many patients might be using it simultaneously without experiencing delays or crashes. It must be built to grow and adapt over time, allowing researchers to add new diagnostic capabilities as medical science discovers additional Parkinson's indicators through ongoing research.

4.3 System requirements

Hardware

- Smartphone or tablet with sufficient memory and storage.
- Touchscreen support with good display quality.
- Microphone for voice input.
- Internet connectivity for updates and data sync.

Software

- Support for Android/iOS platforms.
- Machine learning frameworks for model deployment.
- Secure local and cloud database for data storage.
- Security & Performance
- Encrypted and backed-up data storage.
- Role-based access with authentication.

- Low-latency performance and error monitoring.
- Modular design for easy updates.

4.4 User requirements

Patients

- Easy-to-use interface suitable for elderly and impaired users.
- Clear visual, audio, or voice-assisted instructions.
- Progress tracking with simple charts and reminders for regular use.
- Multilingual support (English, Sinhala, Tamil).

Healthcare Providers

- Access to detailed diagnostic reports with clear explanations.
- Integration with existing hospital/clinical systems.
- Tools to track patient progress over time.

Administrators

- User and access management features.
- System usage analytics and monitoring.
- Strong privacy and security compliance.

4.5 Wireframes (tentative)

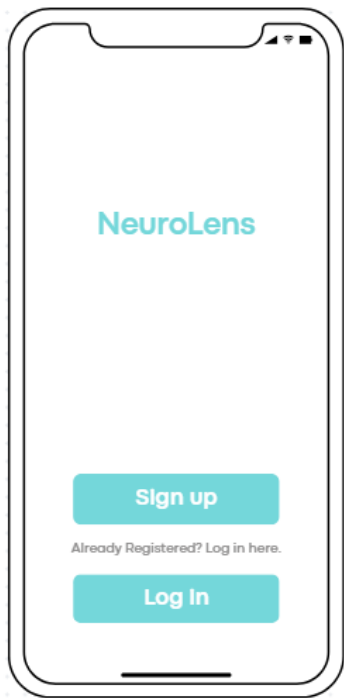


Figure 12:Login / Signup



Figure 11:Recording Screen

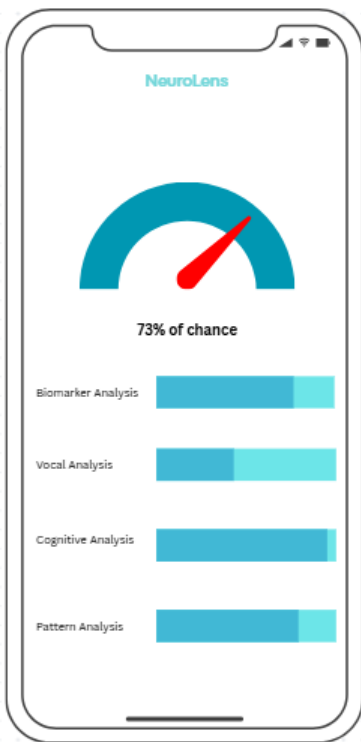


Figure 13:Output Screen

5.COMMERCIALIZATION

Free Tier

The free tier provides access to basic assessments and feedback using selected modules of the app. This tier is ideal for users who want an initial screening or personal monitoring tool. Limitations include restricted access to advanced analytics, multi-modal integration, and real-time AI-powered predictions. Wearable device integration and explainable AI (XAI) outputs are not available in this tier, ensuring the app remains lightweight and accessible while providing essential early detection features.

Paid Tier (LKR 1990)

The paid subscription unlocks the full potential of the app, including complete multi-modal interaction. This tier includes wearable device integration, enabling continuous, real-time monitoring of motor and cognitive signals. This makes the app far more proactive, moving beyond periodic self-assessments to continuous digital biomarker collection. In addition, premium users gain access to explainable AI (XAI) outputs, which highlight how different biomarkers contribute to predictions, along with AI-powered insights and early detection predictions. This tier is designed to provide comprehensive, clinically relevant tools for patients, caregivers, and clinicians.

This tiered approach gives users flexibility according to their needs, supports sustainable app development, and allows continuous improvement. Regular updates will add new features and enhancements, with premium offerings delivering advanced monitoring, interpretability, and integration capabilities.

6.GANTT CHART



Figure 14:Gantt Chart

7. WORK BREAKDOWN CHART

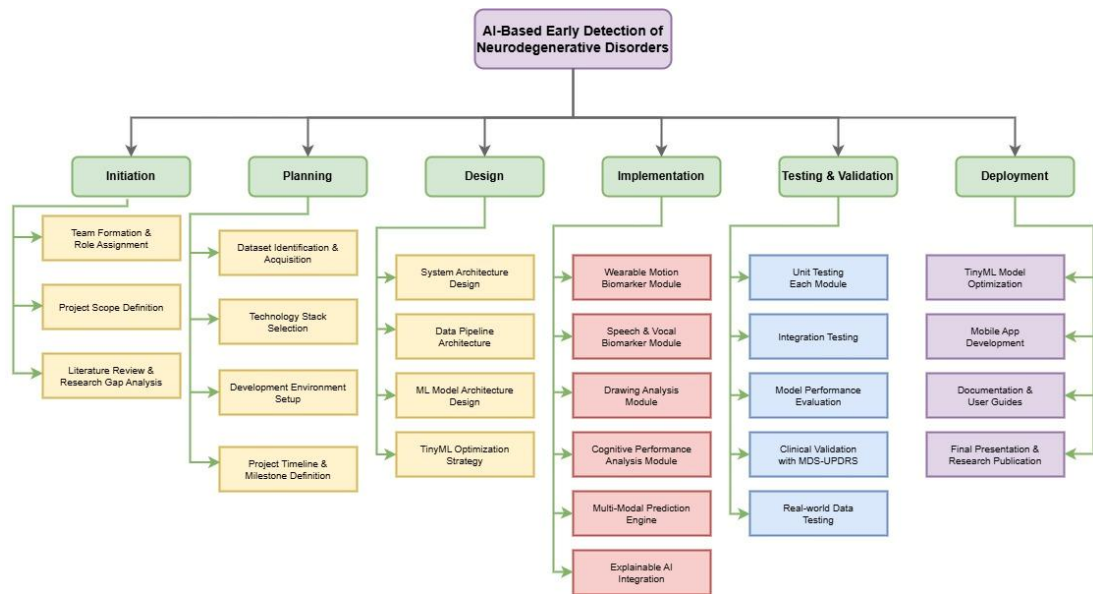
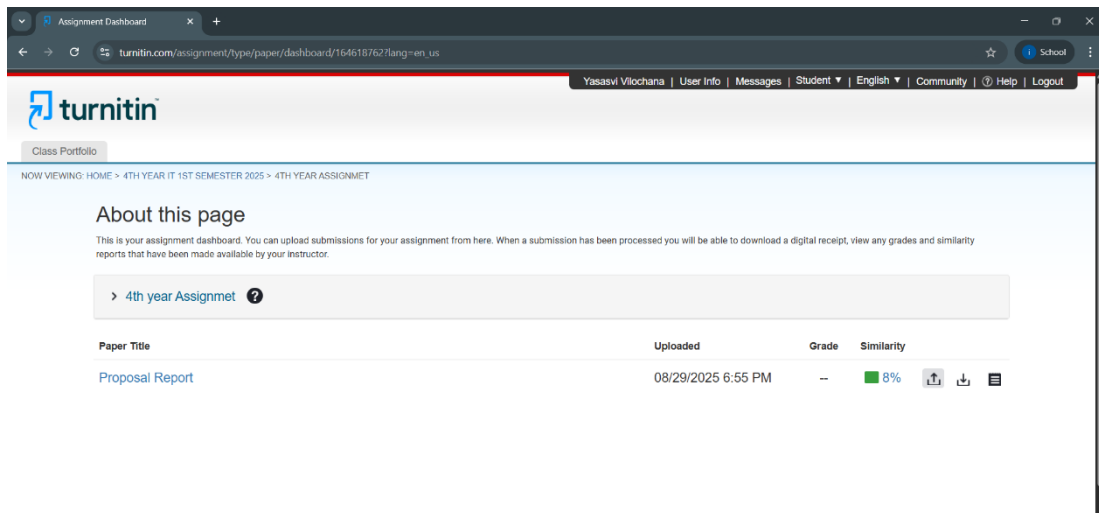


Figure 15: Work Breakdown Structure

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9.APPENDICES



The screenshot displays the Turnitin Assignment Dashboard in a web browser. The browser's address bar shows the URL: turnitin.com/assignment/type/paper/dashboard/164618762?lang=en_us. The Turnitin logo is visible in the top left corner. Below the logo, there is a navigation bar with links: "Class Portfolio", "NOW VIEWING: HOME > 4TH YEAR IT 1ST SEMESTER 2025 > 4TH YEAR ASSIGNMET".

The main content area is titled "About this page" and contains the text: "This is your assignment dashboard. You can upload submissions for your assignment from here. When a submission has been processed you will be able to download a digital receipt, view any grades and similarity reports that have been made available by your instructor."

Below this text is a button labeled "> 4th year Assignmet" with a question mark icon. Below the button is a table with the following data:

Paper Title	Uploaded	Grade	Similarity
Proposal Report	08/29/2025 6:55 PM	—	8%

Figure 16:Appendix